Lab 11: Epigenetics — Gene Regulation

BIOL-1

Name: _____ **Date:** _____

Learning Objectives

By the end of this lab, you will be able to:

1. Explain why gene regulation is essential for multicellular organisms.
2. Describe how the lac operon turns genes on and off in bacteria.
3. Explain how DNA methylation and histone acetylation control gene expression.
4. Describe X-inactivation and its visible effects (calico cats).
5. Give examples of how environment influences gene expression.
6. Model epigenetic switches with a paper game — placing and removing methylation tags — and explain how the same genome can end up with different ON/OFF patterns.

Overview

You have about 20,000 genes — but a brain cell only uses a fraction of them, and a muscle cell uses a different fraction. **Gene regulation** is how cells decide which genes to turn on and which to keep silent. **Epigenetics** takes this one step further: it studies how cells can change gene expression *without changing the DNA sequence itself* — through chemical tags on DNA and its packaging proteins.

In this lab, you will explore how bacteria regulate genes using the **lac operon**, how human cells use **epigenetic mechanisms** like DNA methylation and histone modification, and how the environment can influence which of your genes are active.

Materials

- This worksheet
- Colored pencils (optional)
- A few spare paper scraps for the switchboard game

Part 1: Why Gene Regulation Matters

Think about this: a liver cell and a neuron in your brain have the exact same DNA — the same 20,000 genes. Yet they look completely different, make different proteins, and do completely different jobs. The difference is which genes are turned ON.

1. Do all cells in your body contain the same DNA?

2. Do all cells in your body produce the same proteins?

3. If a muscle cell and a nerve cell have the same DNA, how can they look and function so differently?

Part 2: The Epigenetic Switchboard

In this game, four genes share one DNA strip. Paper tags model epigenetic marks: a MET tag is DNA methylation (gene OFF); a LOOSE tag is open histone packaging (gene ON). Signal slips stand in for life events — the environment, not the DNA sequence, decides which tags get moved.

Step 1 — Draw your DNA strip. On a sheet of paper, draw a long strip divided into four boxes, each box outlined with a **dashed border**. Label them **Gene A (Metabolism)**, **Gene B (Stress response)**, **Gene C (Growth)**, and **Gene D (Pigment)**, and write **ON** in pencil inside every box. Place a **LOOSE** tag on each box (open histones = gene ON); leave the four **MET** tags in a pile beside the strip.

Step 2 — Make your tags and signals. Fold a spare strip into 8 squares and outline each square with a dashed border. Label four squares **MET** and four **LOOSE** in pencil, then carefully tear them apart along the dashed borders — slow, steady tears work best, corner to corner. On five more small scraps, write one signal each: **Healthy diet**, **Chronic stress**, **Toxin exposure**, **Exercise**, **Aging**. Place the signal slips **face-down** on your desk.

Step 3 — Hypothesis first.

4. BEFORE drawing any signals, predict: for each of the five signals, which genes would it tend to silence (add MET) or activate (remove MET / add LOOSE)?

Step 4 — Play three rounds. Each round: place all five signal slips face-down, mix them around, draw one, record it, return it, and mix again before the next round. After each draw, apply its rule and update the pencil ON/OFF labels on your strip, then record all four gene states in the table:

- **Healthy diet:** remove one MET tag (that gene goes back ON).
- **Chronic stress:** remove any MET tag from **Gene B** — stress-response genes fire up under stress — AND add one MET tag to any other gene of your choice (that gene goes OFF).
- **Toxin exposure:** add one MET tag to two different genes of your choice.
- **Exercise:** remove one MET tag AND swap one other gene's tag for a LOOSE tag.
- **Aging:** add one MET tag to one gene of your choice.
  If a signal asks you to remove a MET tag and no gene has one yet, that removal does nothing — record no change for that step and still apply any other part of the rule.

Gene	Start	Round 1	Round 2	Round 3
Gene A (Metabolism)	ON			
Gene B (Stress response)	ON			
Gene C (Growth)	ON			
Gene D (Pigment)	ON			

Step 5 — Twin connection.

5. Identical twins start with the same DNA but can end up with different traits. Imagine a classmate playing the same game with the same four genes: using your switchboard, explain how the SAME genome could end up with a different ON/OFF pattern than yours.

Part 3: The Lac Operon — Gene Regulation in Bacteria

E. coli bacteria live in your intestines. They can digest lactose (milk sugar), but making the enzymes to do so costs energy. Why make enzymes you don't need? The lac operon is an elegant ON/OFF switch that solves this problem.

How the lac operon works:

- **When lactose is ABSENT:** The repressor protein binds to the operator (blocks the road). RNA polymerase cannot get through. The enzyme genes stay OFF. No energy wasted.
- **When lactose is PRESENT:** Lactose binds to the repressor and changes its shape. The repressor falls off the operator. RNA polymerase reads the genes. Enzymes are made. Lactose gets digested.

6. What is the job of the repressor protein in the lac operon?

7. What happens to the repressor protein when lactose enters the cell?

8. Why is the lac operon system energy-efficient for the bacterium?

9. Prediction: If a mutation destroys the repressor gene (so no repressor is made), will the lac operon genes be always ON, always OFF, or normal?

10. Prediction: If a mutation in the promoter prevents RNA polymerase from binding, will the genes be always ON, always OFF, or normal?

Part 4: Epigenetic Mechanisms

Epigenetics literally means "above genetics." These are chemical modifications that sit ON TOP of DNA and histones and control whether genes are accessible or locked away — all without changing a single base in the DNA sequence.

11. What is DNA methylation? Does it typically turn genes ON or OFF?

12. What is histone acetylation? Does it typically turn genes ON or OFF?

13. Fill in the table (circle the correct answers):

Chromatin Type	DNA Packaging	Gene Expression
Euchromatin	Loose / Tight	ON / OFF
Heterochromatin	Loose / Tight	ON / OFF

14. In a skin cell, the gene for keratin (a skin protein) is actively making protein. Is the keratin gene packaged as euchromatin or heterochromatin in this cell?

15. In that SAME skin cell, the gene for hemoglobin (a blood protein) is completely silent. Is the hemoglobin gene packaged as euchromatin or heterochromatin?

16. A cell that performs histone acetylation on a gene is essentially doing what to that gene's expression?

Part 5: X-Inactivation — Epigenetics You Can See

Female mammals have two X chromosomes (XX), while males have one (XY). To balance gene dosage, one X chromosome in each female cell is randomly shut down early in development. The result? Patches of cells using Mom's X and patches using Dad's X.

17. What is X-inactivation?

18. What is the condensed, inactivated X chromosome called?

19. Calico cats have patches of orange and black fur. The gene for coat color is on the X chromosome. Using what you know about X-inactivation, explain why calico cats have a patchy pattern instead of a uniform color.

20. Why are calico cats almost always female?

Part 6: Environment and Epigenetics

Your genes are not your destiny. While you cannot change your DNA sequence, your lifestyle and environment can influence which genes are active through epigenetic modifications.

21. Name two environmental factors that can change gene expression through epigenetics, and briefly explain how:

1.

2.

22. Identical twins share 100% of their DNA. Yet as they age, they often develop different diseases, different body types, and even different personalities. Using what you learned in this lab — including your Part 2 switchboard — explain how this is possible.

23. A cancer researcher finds that a patient's tumor suppressor gene is heavily methylated but has NO mutations in its DNA sequence. Could this contribute to cancer? Explain.